

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – EXPLORATORY DATA ANALYSIS

Course Code:	DSA05	Course Title:	Query Processing for Data Science in Real Time Applications
CO Assessed:	CO4	Assessment:	Tool 1 – Exploratory Data Analysis
Weightage:		Total Marks:	25 Marks
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.. (BL4 and BL5)			
Student Name:	U.Ananya	Reg. No.:	192424382
Section / Batch:	AI&DS / 24 Batch	Date of Submission:	19-08-2026
Faculty In-charge:	K. Veena Devi	Assessment Mode:	Individual Submission

Code of Conduct

I __U.Ananya(192424382)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: _____U.Ananya_____

Dataset Exploration (05 Marks)	Data Cleaning and Preparation (05 Marks)	Statistical Analysis (05 Marks)	Data Visualization (05 Marks)	Insights, Interpretation and Reporting (05 Marks)	Total (25 Marks)

1. Titanic Passenger Survival Analysis

Problem Statement

The aim of this project is to analyze the Titanic passenger dataset and identify the factors that were associated with passenger survival. The analysis includes handling missing values, studying survival based on gender, age, and passenger class, detecting outliers, and visualizing the data.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("titanic.csv")
print(df.head())
print(df.isnull().sum())

df["Age"] = df["Age"].fillna(df["Age"].median())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
df["Fare"] = df["Fare"].fillna(df["Fare"].median())

print("\nSurvival by Gender:")
print(df.groupby("Sex")["Survived"].mean())

print("\nSurvival by Class:")
print(df.groupby("Pclass")["Survived"].mean())

print("\nAverage Age of Survivors:")
print(df.groupby("Survived")["Age"].mean())

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
sns.boxplot(y=df["Age"])
plt.title("Age Outliers")

plt.subplot(1, 2, 2)
sns.boxplot(y=df["Fare"])
plt.title("Fare Outliers")
```

```

plt.show()

sns.countplot(x="Sex", hue="Survived", data=df)

plt.title("Survival by Gender")

plt.show()

sns.histplot(df["Age"], bins=20, kde=True)

plt.title("Age Distribution")

plt.show()

sns.barplot(x="Pclass", y="Survived", data=df)

plt.title("Survival Rate by Passenger Class")

plt.show()

```

Output:

```

plt.show()

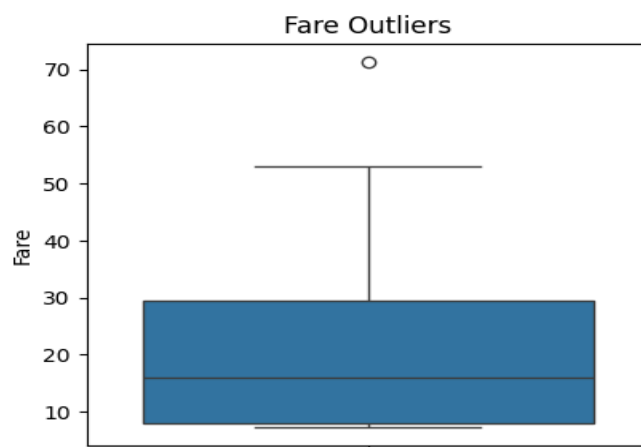
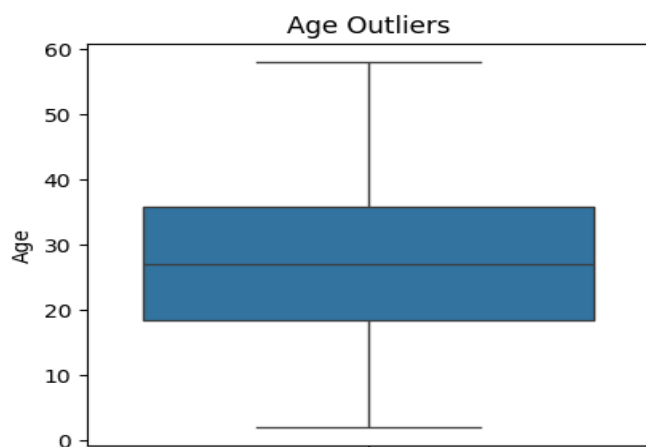
... PassengerId  Survived  Pclass   Sex   Age   Fare Embarked
0         1         0         3   male  22.0   7.2500      S
1         2         1         1  female  38.0  71.2833      C
2         3         1         3  female  26.0   7.9250      S
3         4         1         1  female  35.0  53.1000      S
4         5         0         3   male  35.0   8.0500      S
PassengerId    0
Survived        0
Pclass         0
Sex            0
Age           1
Fare          1
Embarked       0
dtype: int64

Survival by Gender:
Sex
female    0.9
male      0.1
Name: Survived, dtype: float64

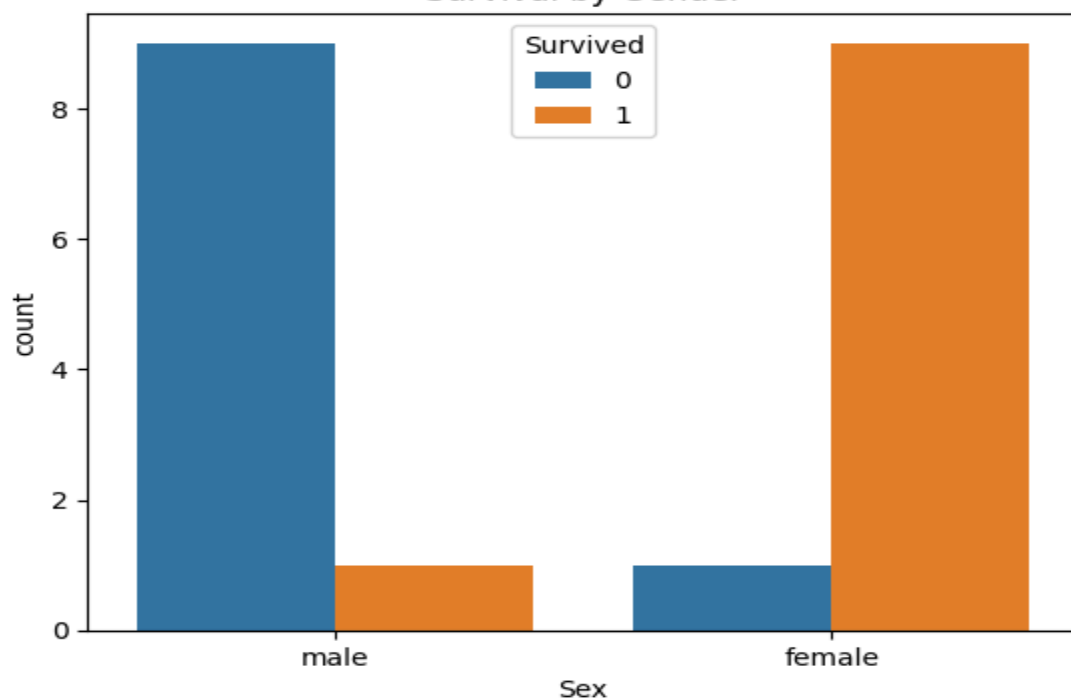
Survival by Class:
Pclass
1    0.833333
2    0.666667
3    0.272727
Name: Survived, dtype: float64

Average Age of Survivors:
Survived
0    24.6
1    30.7
Name: Age, dtype: float64

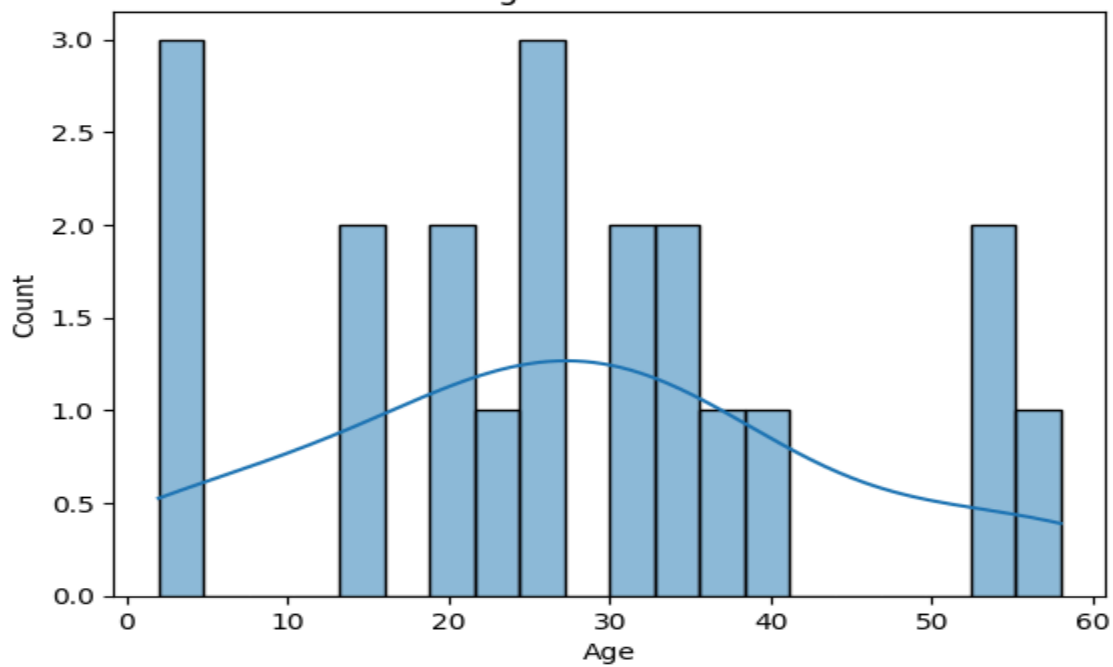
```

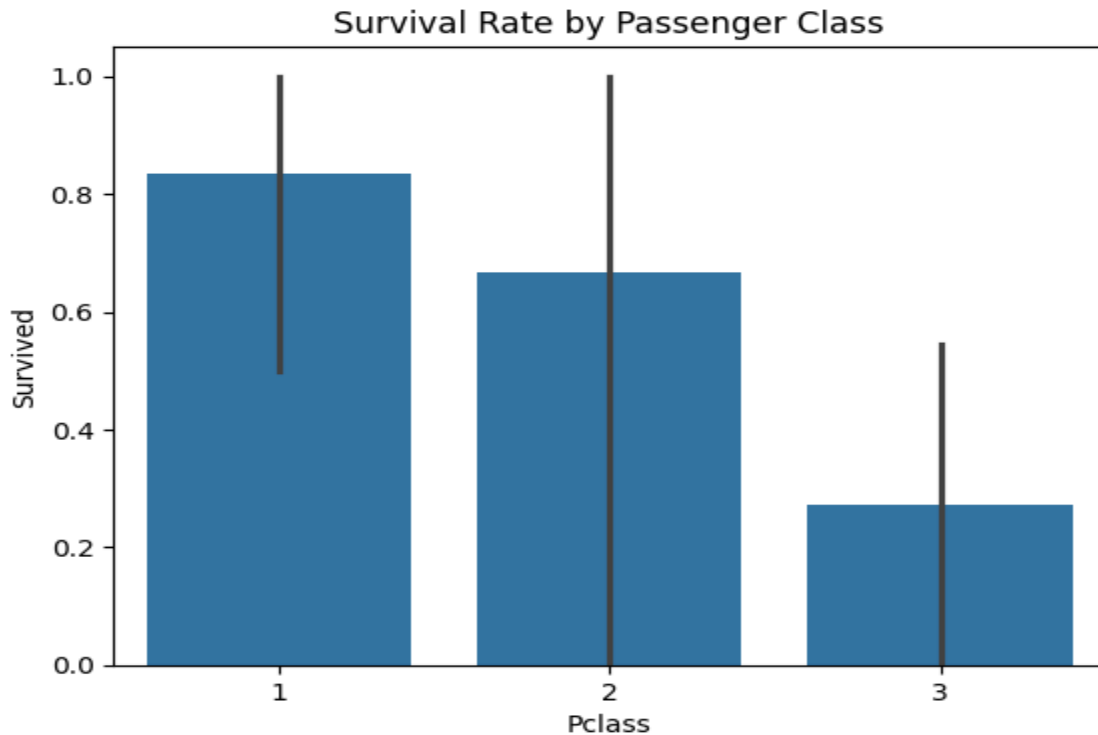


Survival by Gender



Age Distribution





Key Findings

- Female passengers had a higher survival rate than male passengers.
- First-class passengers generally had a better chance of survival than second- and third-class passengers.
- Younger passengers had different survival patterns compared with older passengers.
- Age and Fare contain outliers, which can be identified using boxplots.
- Missing values in Age, Fare, and Embarked were handled using median or mode values.
- Overall, gender and passenger class were important factors associated with survival.

Conclusion

The Titanic dataset shows that passenger survival was influenced by factors such as gender, age, and passenger class. Data cleaning and visualization helped identify these patterns and made the dataset easier to analyze.

2. Employee Attrition Analysis

Problem Statement

The aim of this project is to analyze the IBM HR Employee Attrition dataset and identify the major factors that influence employee attrition. The analysis focuses on employee salary, job role, years at the company, and other features to understand why employees may leave an organization.

Python Code

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv("IBM_HR_Employee_Attrition.csv")

print(df.head())

print("\nAttrition Distribution:")

print(df["Attrition"].value_counts())

sns.countplot(x="Attrition", data=df)

plt.title("Employee Attrition")

plt.show()

sns.boxplot(x="Attrition", y="MonthlyIncome", data=df)

plt.title("Salary vs Attrition")

plt.show()

plt.figure(figsize=(10,5))

sns.countplot(x="JobRole", hue="Attrition", data=df)

plt.xticks(rotation=45)

plt.title("Job Role vs Attrition")

plt.show()
```

```
sns.boxplot(x="Attrition", y="YearsAtCompany", data=df)
```

```
plt.title("Years at Company vs Attrition")
```

```
plt.show()
```

```
features = ["Age", "MonthlyIncome", "YearsAtCompany", "JobSatisfaction"]
```

```
sns.heatmap(df[features].corr(), annot=True, cmap="coolwarm")
```

```
plt.title("Employee Feature Relationships")
```

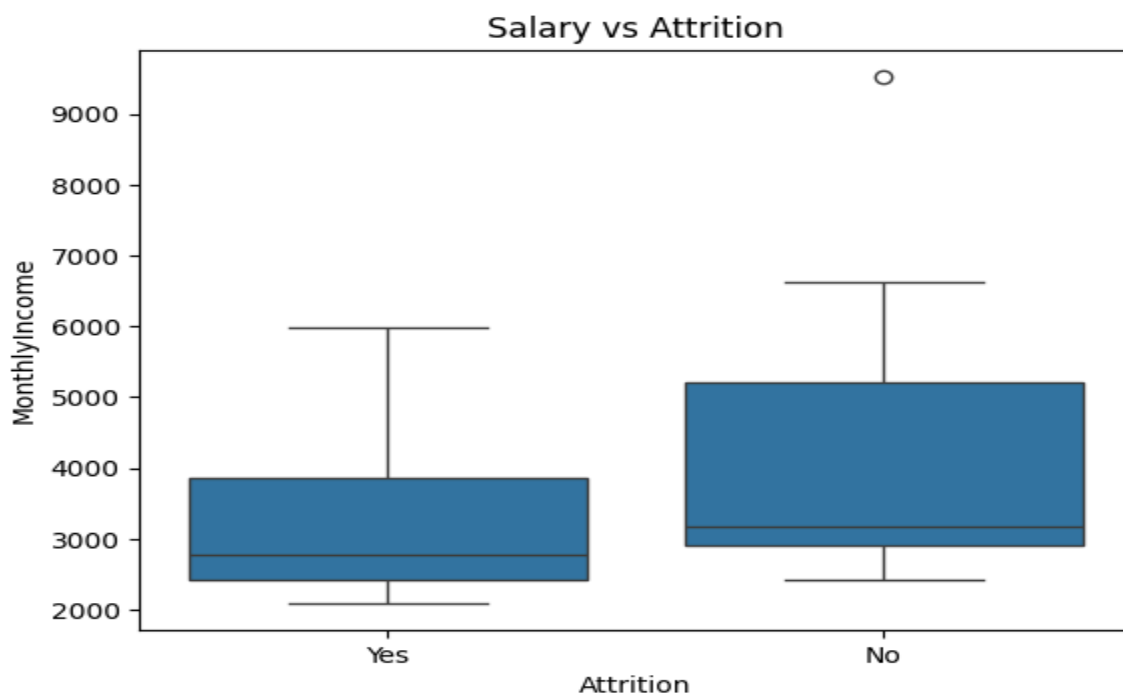
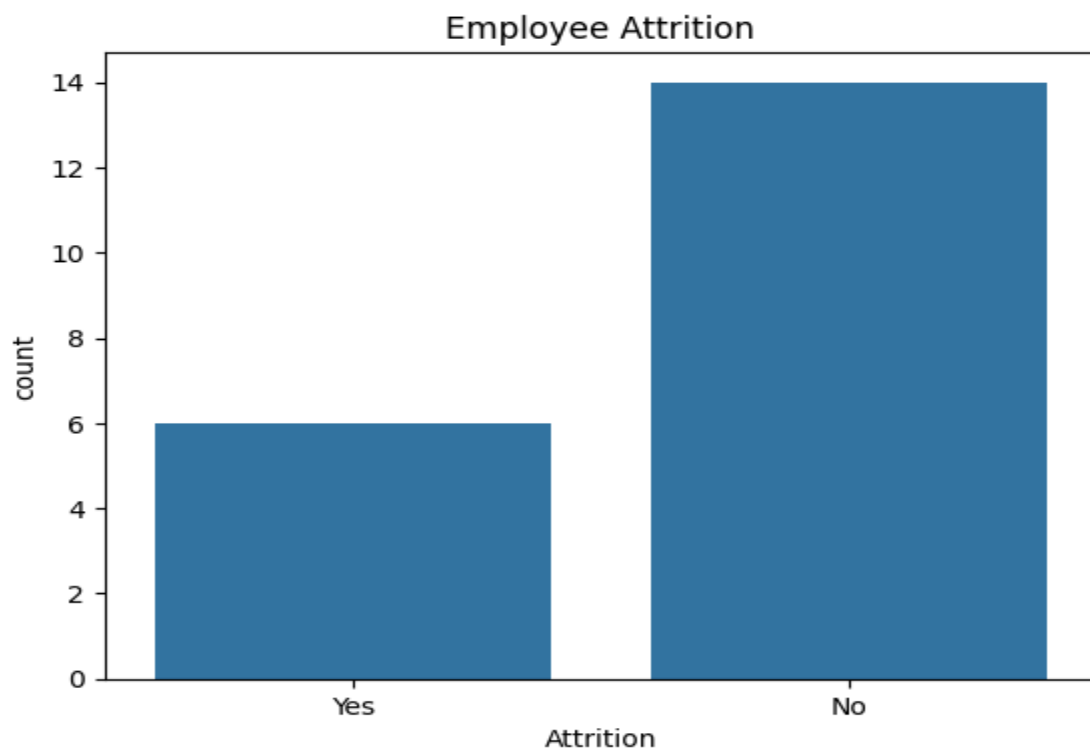
```
plt.show()
```

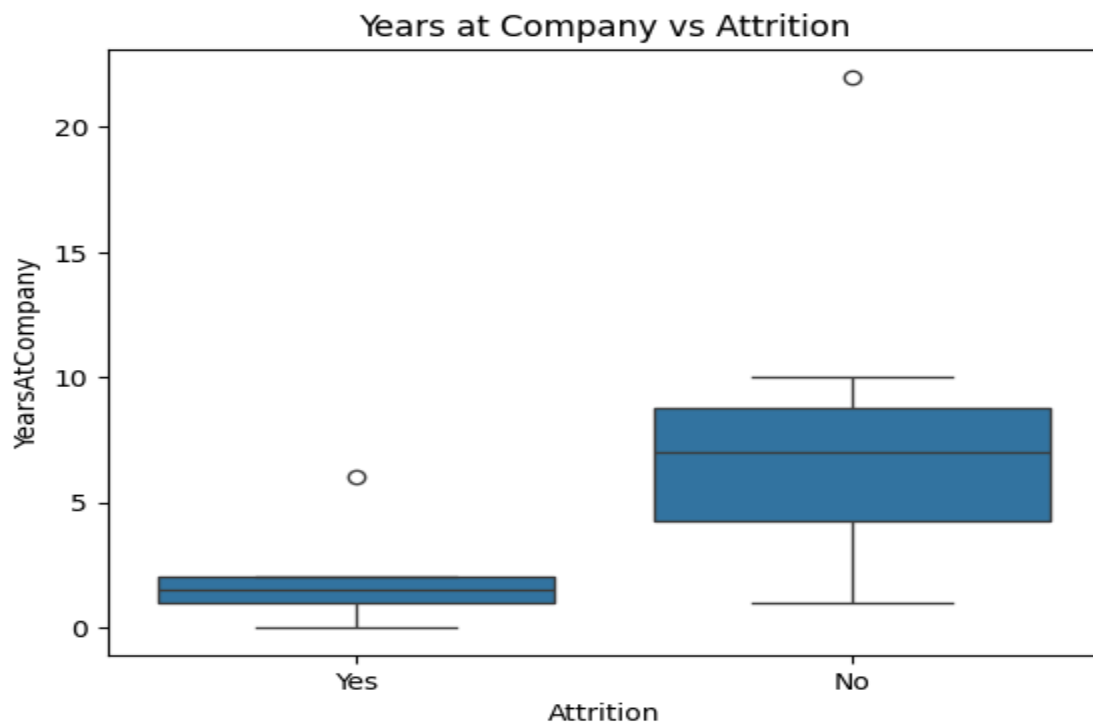
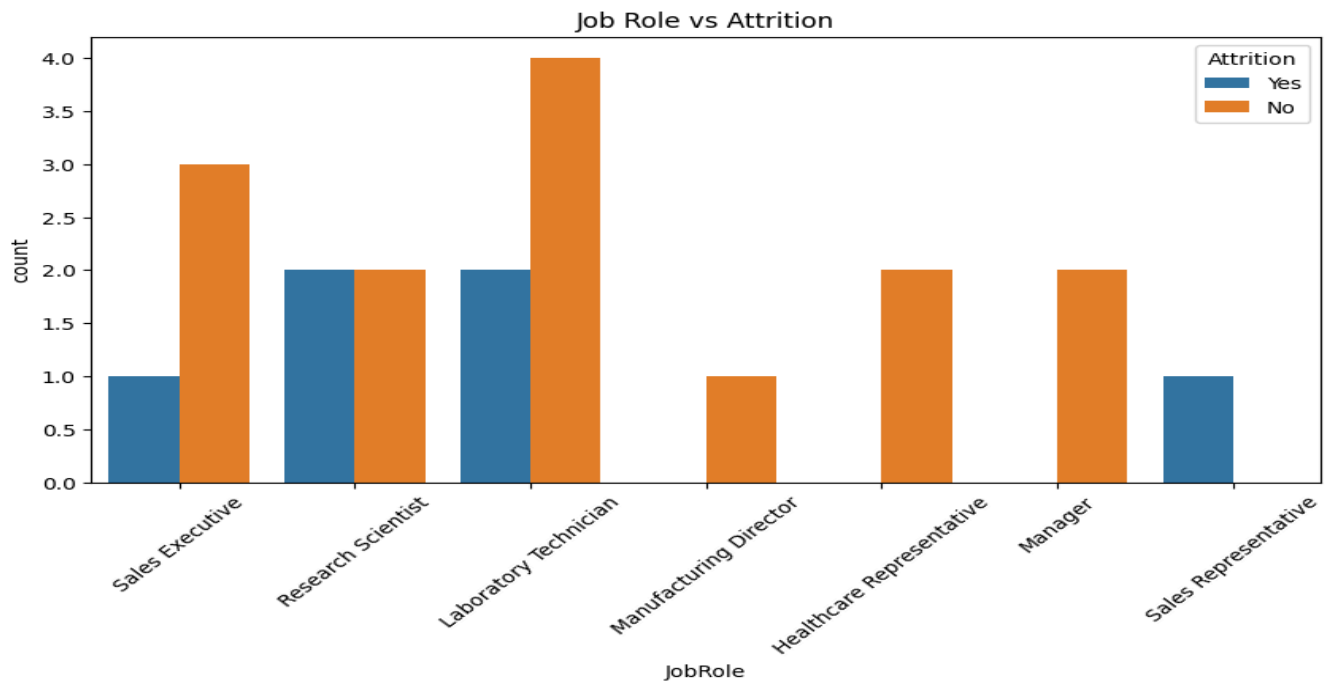
Output:

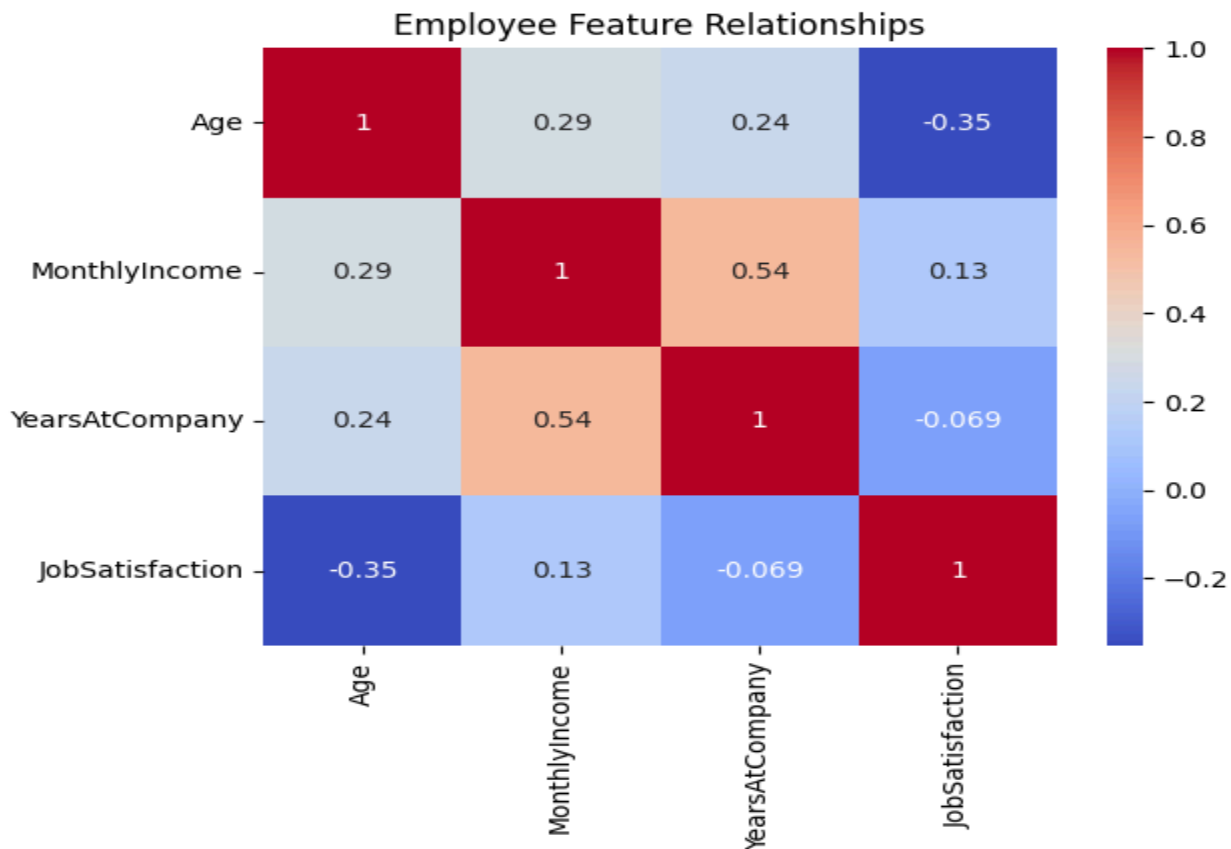
```
...   Age  MonthlyIncome  JobRole  YearsAtCompany  Attrition  \
0    41         5993   Sales Executive           6         Yes
1    49         5130  Research Scientist          10         No
2    37         2090  Laboratory Technician           0         Yes
3    33         2909  Research Scientist           8         No
4    27         3468  Laboratory Technician           2         No

   JobSatisfaction
0                4
1                2
2                3
3                3
4                4

Attrition Distribution:
Attrition
No      14
Yes      6
Name: count, dtype: int64
```







Key Findings

- The dataset contains both employees who stayed and employees who left the company.
- Employees with lower salaries may show higher attrition compared with higher-paid employees.
- Attrition can vary across different job roles.
- Employees with fewer years at the company may have a higher tendency to leave.
- Factors such as age, income, job satisfaction, and experience can be related to employee attrition.
- Visualizations make it easier to identify patterns and compare employees who stayed with those who left.

Recommendations

The organization can reduce employee attrition by:

- Improving salary and benefits where necessary.
- Providing better career growth and promotion opportunities.
- Improving employee job satisfaction and work environment.
- Providing training and development programs.
- Identifying employees with high attrition risk and taking early supportive action.

Conclusion

Employee attrition analysis helps organizations understand why employees leave and which factors are associated with turnover. By analyzing salary, job role, experience, satisfaction, and other employee characteristics, companies can develop better retention strategies and create a more supportive workplace.

3. Housing Price Analysis

Problem Statement

The aim of this project is to analyze a House Prices dataset and identify the major factors that affect property prices. The analysis focuses on numerical and categorical features such as area, number of bedrooms, bathrooms, furnishing status, and price.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("Housing.csv")
print(df.head())
print(df.info())
print(df.describe())
sns.histplot(df["price"], kde=True)
plt.title("House Price Distribution")
plt.show()
sns.boxplot(y=df["price"])
plt.title("House Price Outliers")
plt.show()
sns.scatterplot(x="area", y="price", data=df)
plt.title("Area vs House Price")
plt.show()
sns.boxplot(x="bedrooms", y="price", data=df)
```

```

plt.title("Bedrooms vs House Price")

plt.show()

num_cols = ["area", "bedrooms", "bathrooms", "stories", "parking", "price"]

plt.figure(figsize=(8,6))

sns.heatmap(df[num_cols].corr(), annot=True, cmap="coolwarm")

plt.title("Correlation Heatmap")

plt.show()

sns.boxplot(x="furnishingstatus", y="price", data=df)

plt.xticks(rotation=20)

plt.title("Furnishing Status vs Price")

plt.show()

```

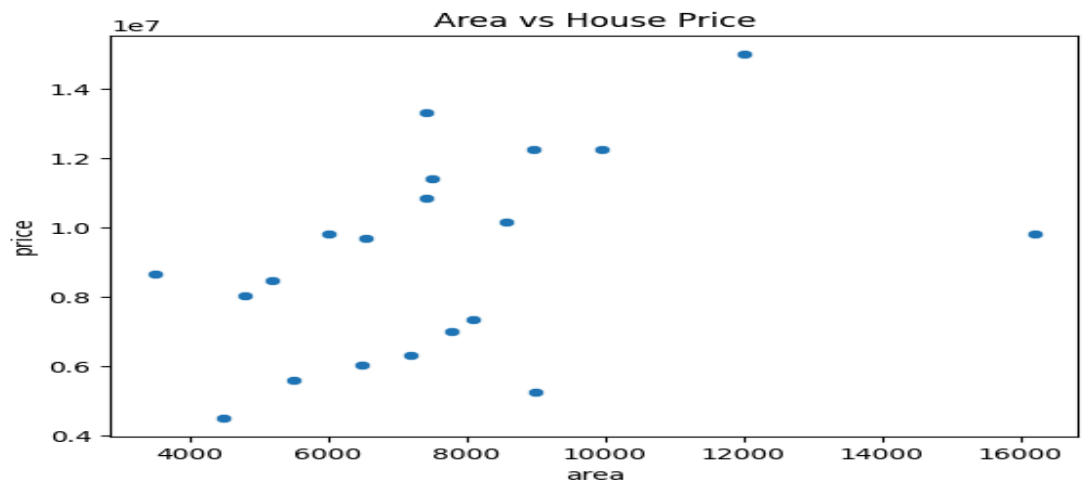
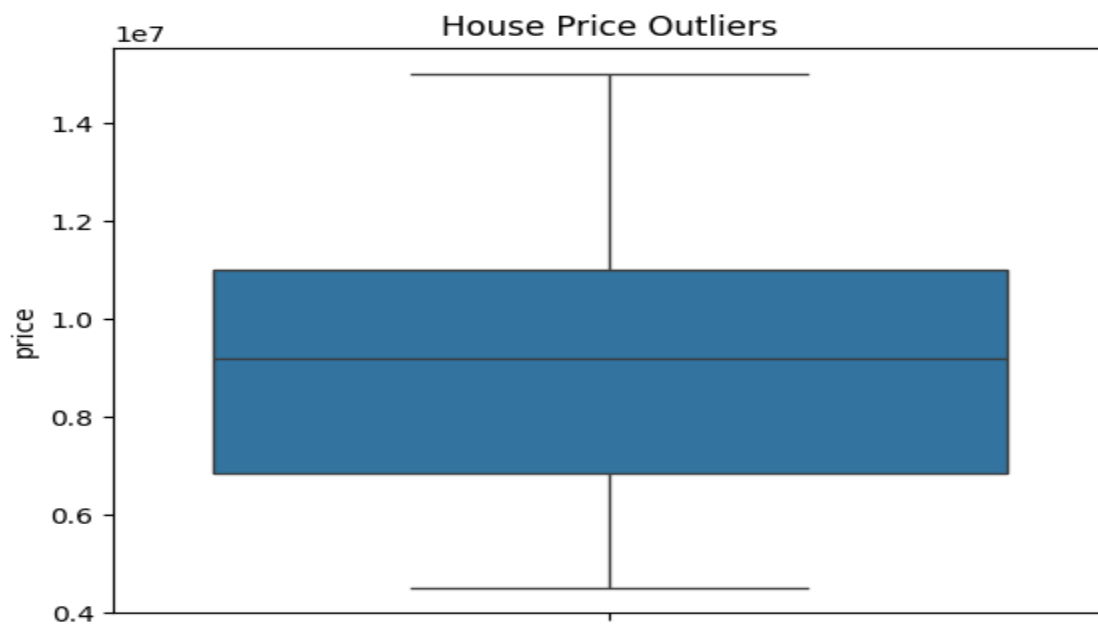
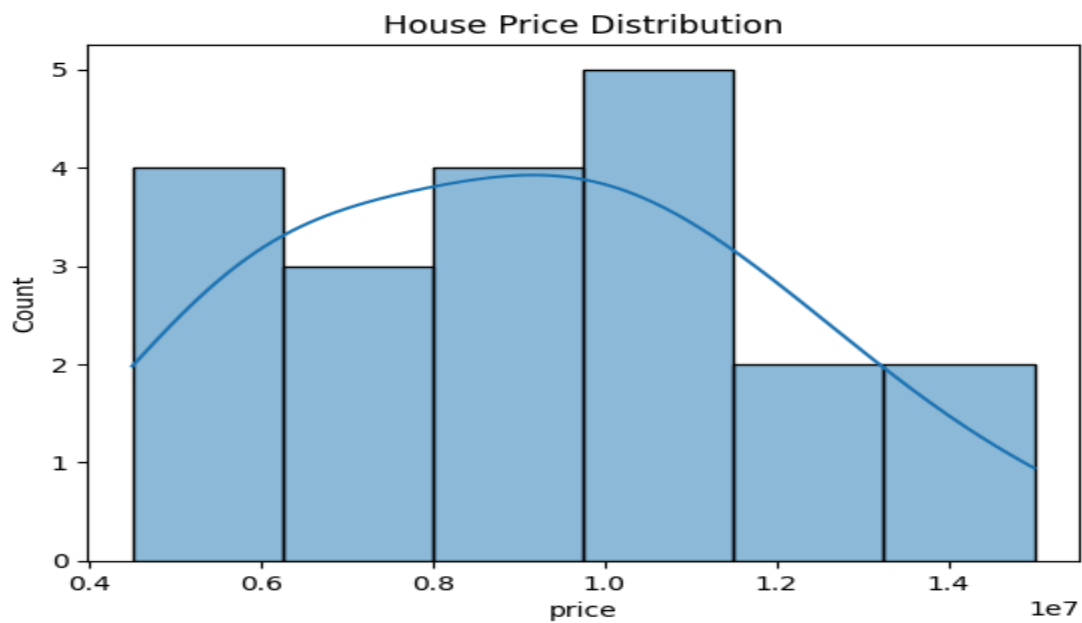
Output:

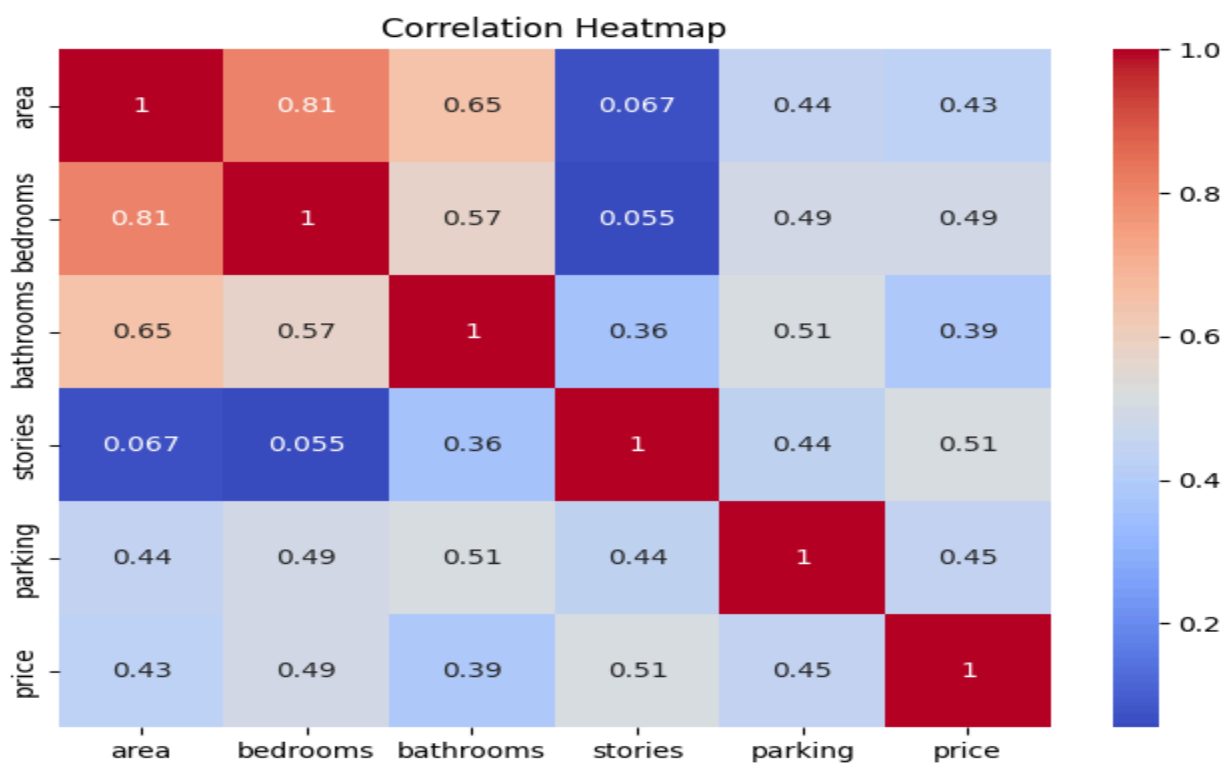
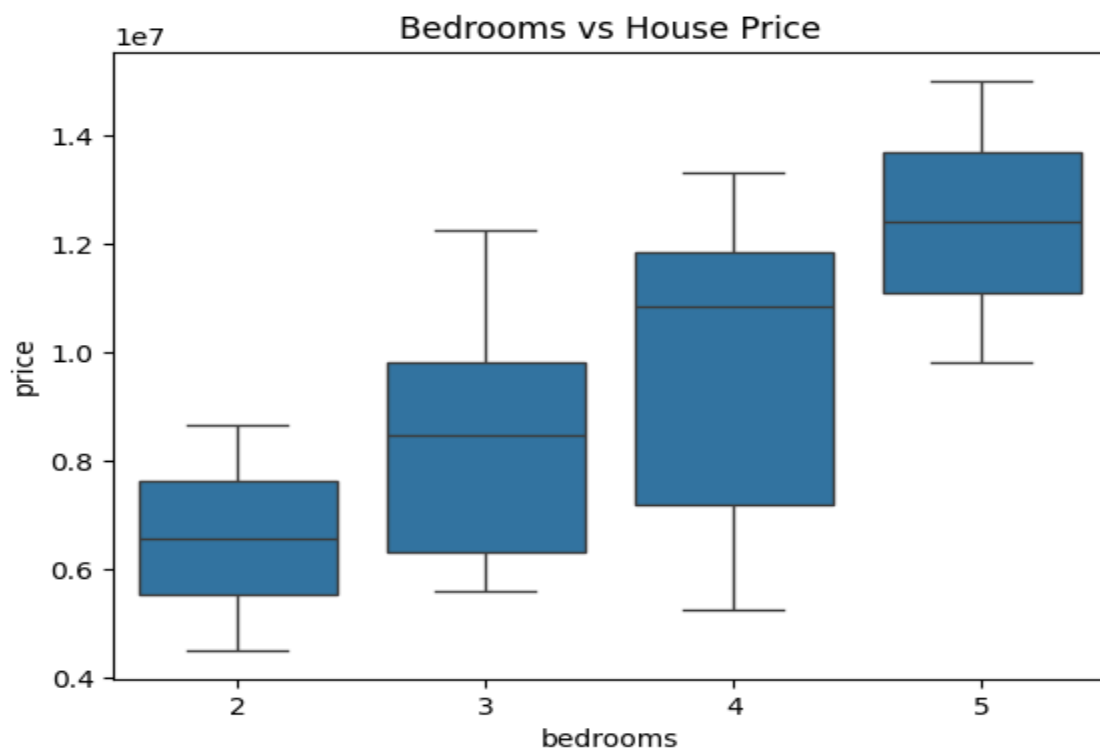
```

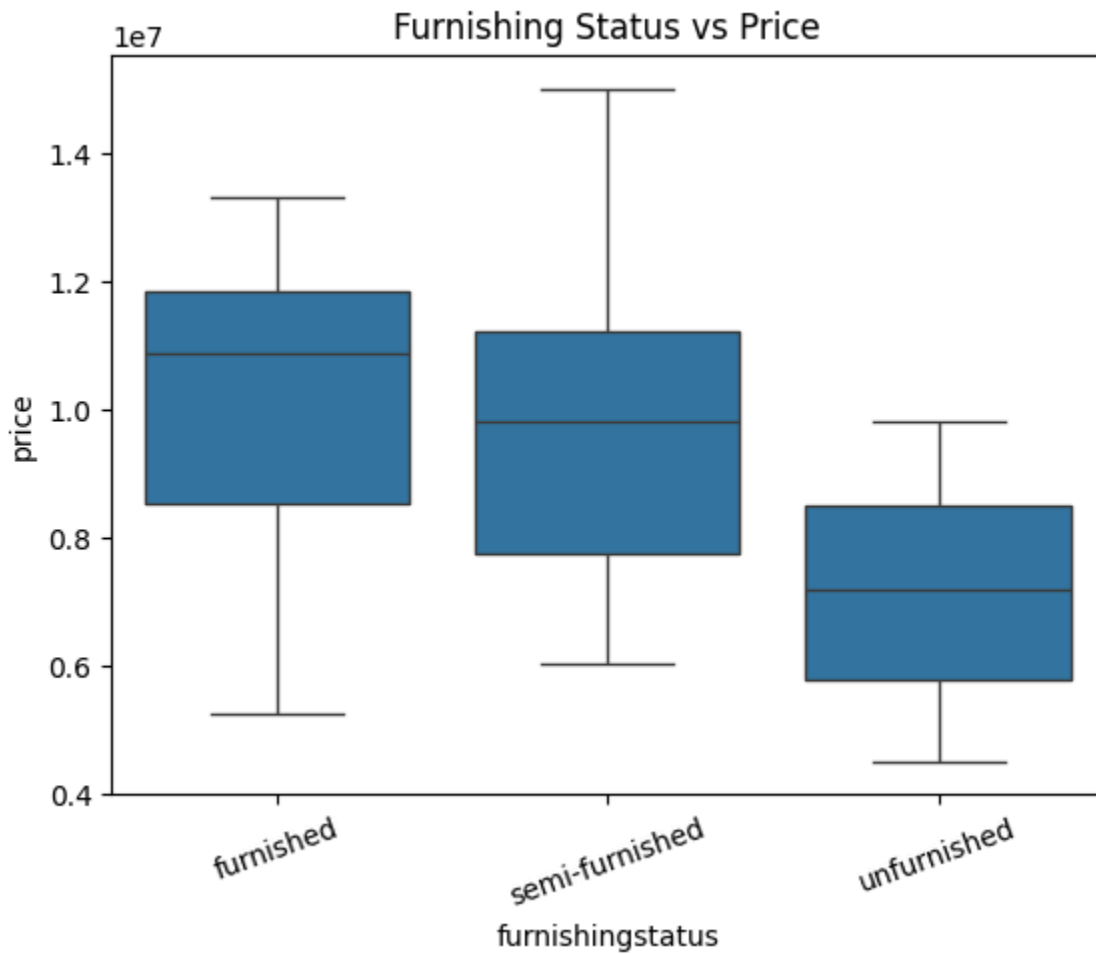
   area  bedrooms  bathrooms  stories  parking  furnishingstatus  price
0  7420         4          2        3         2      furnished  13300000
1  8960         4          4        4         3      furnished  12250000
2  9960         3          2        2         2  semi-furnished  12250000
3  7500         4          2        2         3      furnished  11410000
4  7420         4          1        2         2      furnished  10850000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   area                  20 non-null    int64
1   bedrooms              20 non-null    int64
2   bathrooms             20 non-null    int64
3   stories               20 non-null    int64
4   parking               20 non-null    int64
5   furnishingstatus      20 non-null    object
6   price                 20 non-null    int64
dtypes: int64(6), object(1)
memory usage: 1.2+ KB
None

```

	area	bedrooms	bathrooms	stories	parking	price
count	20.000000	20.000000	20.000000	20.000000	20.000000	2.000000e+01
mean	7634.500000	3.450000	2.100000	2.350000	1.700000	9.083100e+06
std	2842.007213	0.825578	0.788069	0.988087	0.864505	2.882519e+06
min	3500.000000	2.000000	1.000000	1.000000	0.000000	4.500000e+06
25%	5875.000000	3.000000	2.000000	2.000000	1.000000	6.825000e+06
50%	7420.000000	3.000000	2.000000	2.000000	2.000000	9.163000e+06
75%	8675.000000	4.000000	2.250000	3.000000	2.000000	1.099000e+07
max	16200.000000	5.000000	4.000000	4.000000	3.000000	1.500000e+07







Key Findings

- Area generally has a strong relationship with house price. Larger houses tend to have higher prices.
- The number of bedrooms and bathrooms can also influence property prices.
- The boxplot helps identify very high-priced or low-priced properties as outliers.
- The correlation heatmap shows the strength of relationships between numerical features.
- Categorical factors such as furnishing status can also contribute to differences in house prices.

Major Pricing Factors

The important factors affecting house prices are generally:

1. Area of the house
2. Number of bedrooms
3. Number of bathrooms

4. Number of stories
5. Parking availability
6. Furnishing status

Conclusion

The analysis shows that housing prices are influenced by several property characteristics. Area, bedrooms, bathrooms, and other facilities play an important role in determining the price of a house. Data visualization and correlation analysis help identify these relationships and provide a better understanding of housing market trends.

4. COVID-19 Data Exploration

Problem Statement

The aim of this project is to analyze COVID-19 cases across different countries and identify important trends in cases, deaths, and recoveries. The analysis uses statistical calculations and visualizations to understand the spread and impact of COVID-19.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("covid19.csv")
print(df.head())
print("\nSummary Statistics:")
print(df.describe())
print("\nTotal Cases:", df["Confirmed"].sum())
print("Total Deaths:", df["Deaths"].sum())
print("Total Recoveries:", df["Recovered"].sum())
top = df.groupby("Country")["Confirmed"].sum().sort_values(ascending=False).head(10)
print("\nTop 10 Countries:")
print(top)
top.plot(kind="bar", figsize=(10,5))
```



```

plt.title("Top 10 Countries by Confirmed Cases")

plt.xlabel("Country")

plt.ylabel("Confirmed Cases")

plt.xticks(rotation=45)

plt.show()

df["Date"] = pd.to_datetime(df["Date"])

daily = df.groupby("Date")[["Confirmed", "Deaths", "Recovered"]].sum()

daily.plot(figsize=(10,5))

plt.title("COVID-19 Trends Over Time")

plt.xlabel("Date")

plt.ylabel("Number of Cases")

plt.show()

death_rate = (df["Deaths"].sum() / df["Confirmed"].sum()) * 100

print("\nOverall Death Rate:", round(death_rate, 2), "%")

```

Output:

```

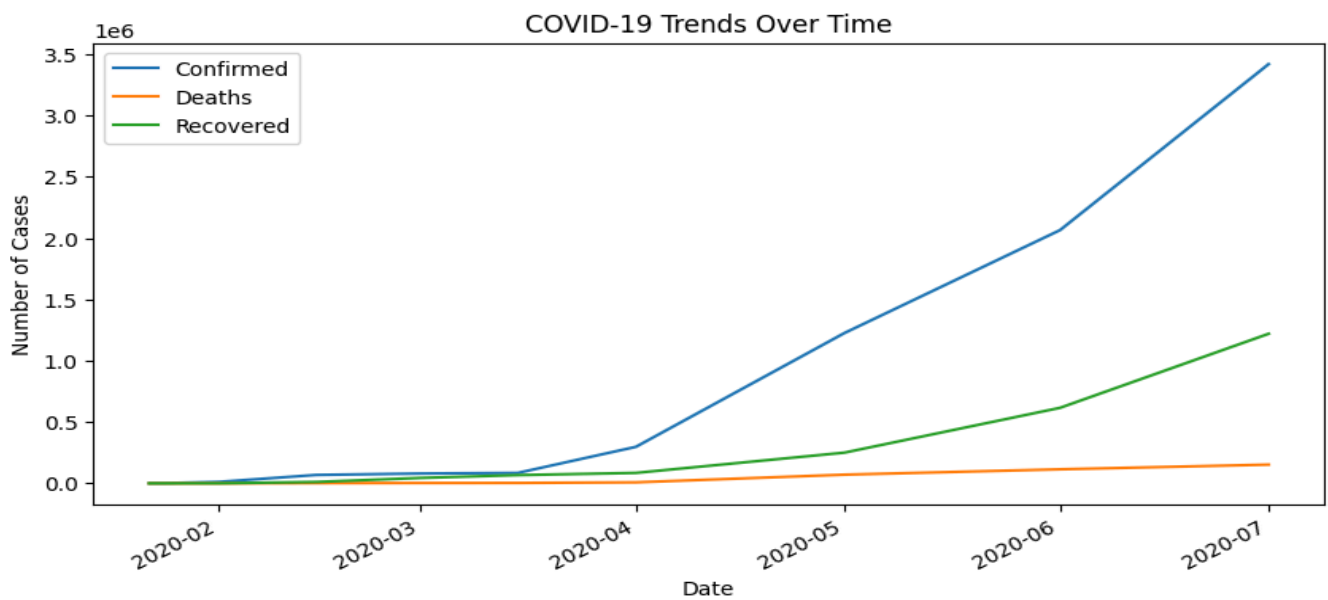
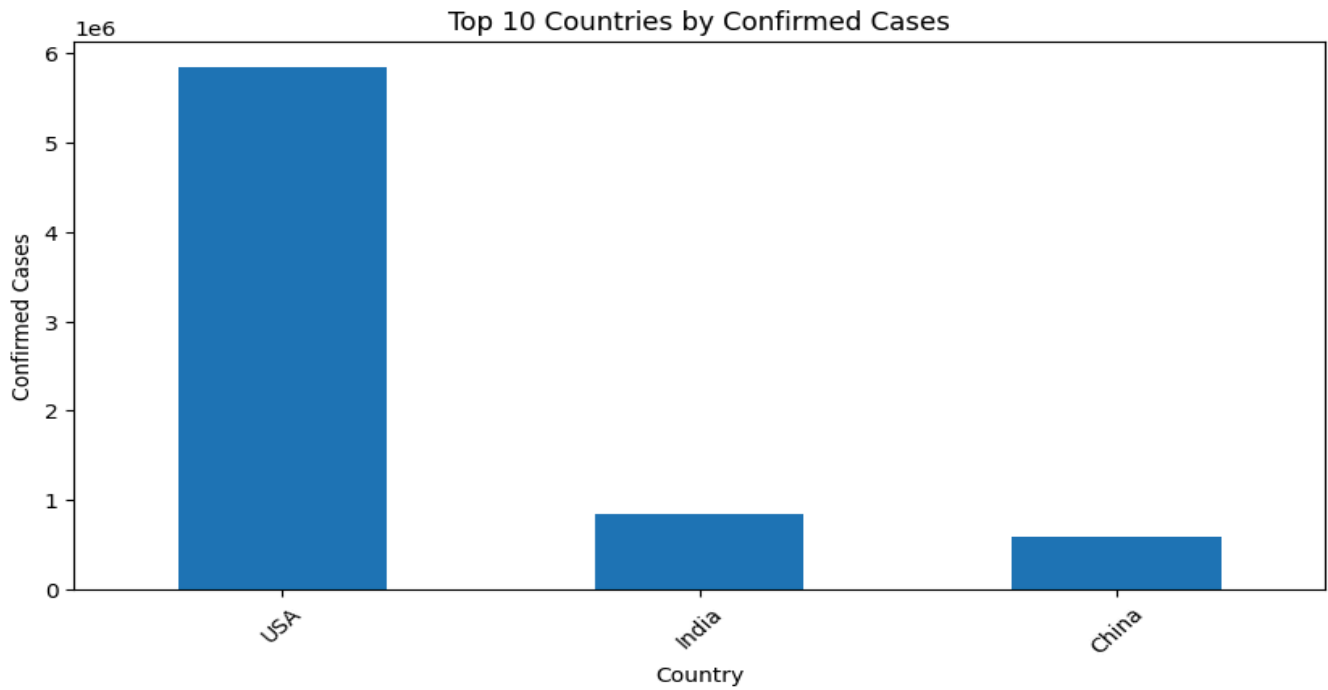
...
   Date Country  Confirmed  Deaths  Recovered
0 2020-01-22  China         548        17         28
1 2020-01-22  India          0          0          0
2 2020-01-22   USA           1          0          0
3 2020-01-25  China       1975         56         49
4 2020-01-25  India          0          0          0

Summary Statistics:
              Confirmed              Deaths              Recovered
count  3.000000e+01      30.000000      30.000000
mean   2.418715e+05     11800.966667     76461.733333
std    6.051350e+05     31259.324756    169141.714861
min    0.000000e+00          0.000000          0.000000
25%    9.000000e+00          0.000000          0.000000
50%    7.787000e+03       164.000000       195.500000
75%    8.414600e+04      4639.000000     77291.250000
max    2.728800e+06    130105.000000    781970.000000

Total Cases: 7256146
Total Deaths: 354029
Total Recoveries: 2293852

Top 10 Countries:
Country
USA      5844466
India     832913
China     578767
Name: Confirmed, dtype: int64

```



Key Findings

- Summary statistics help understand the overall number of confirmed cases, deaths, and recoveries.
- Some countries recorded significantly higher numbers of confirmed cases than others.
- The comparison between confirmed cases, deaths, and recoveries helps understand the overall impact of the pandemic.
- The time-series graph shows how COVID-19 cases changed over time.

- Countries with high infection numbers require greater attention to healthcare capacity and preventive measures.

Actionable Insights

- Governments can use case trends to plan healthcare resources and medical facilities.
- High-risk regions can be identified using infection and death rates.
- Historical trends can help authorities prepare for possible increases in cases.
- Data visualization makes it easier to communicate important pandemic trends.

Conclusion

COVID-19 data analysis helps identify the spread, severity, and progression of the pandemic across countries and over time. Statistical analysis and visualization provide useful insights that can support public-health planning and resource management.

5. Automobile Market Analysis

Problem Statement

The aim of this project is to analyze an Automobile MPG dataset and understand the factors that affect vehicle fuel efficiency. The analysis focuses on MPG, engine size, vehicle weight, and vehicle type.

Python Code

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_csv("auto_mpg.csv")

print(df.head())

print(df.describe())

sns.histplot(df["mpg"], kde=True)

plt.title("MPG Distribution")
```

```
plt.show()
```

```
sns.boxplot(x="vehicle_type", y="mpg", data=df)
```

```
plt.title("Mileage by Vehicle Type")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

```
sns.scatterplot(x="engine_size", y="mpg", data=df)
```

```
plt.title("Engine Size vs MPG")
```

```
plt.show()
```

```
sns.scatterplot(x="weight", y="mpg", data=df)
```

```
plt.title("Vehicle Weight vs MPG")
```

```
plt.show()
```

```
sns.boxplot(y=df["mpg"])
```

```
plt.title("MPG Outliers")
```

```
plt.show()
```

```
print("\nCorrelation:")
```

```
print(df[["mpg", "engine_size", "weight"]].corr())
```

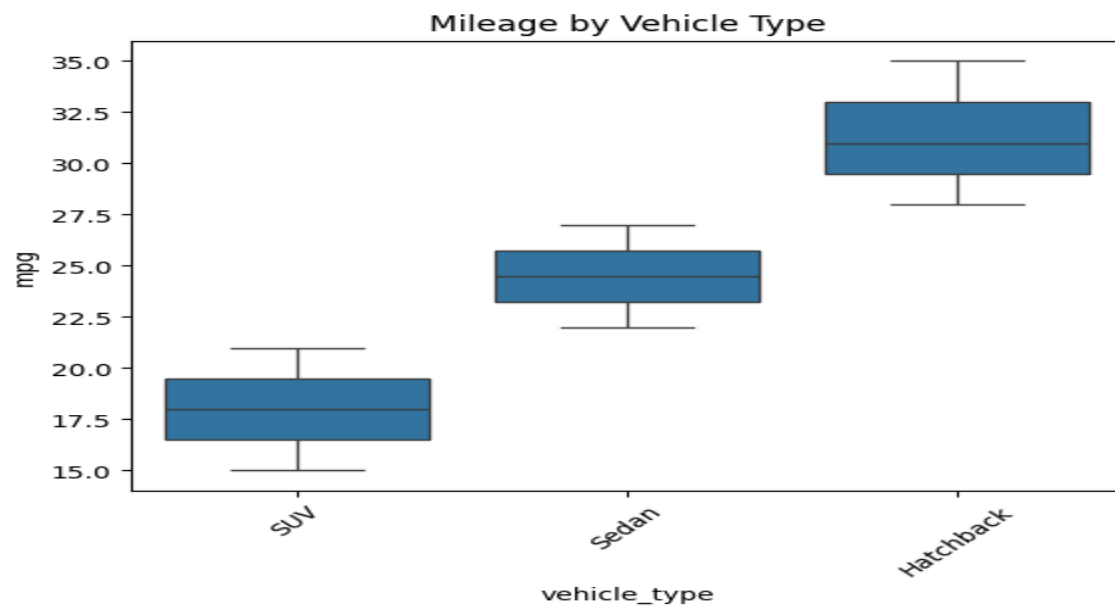
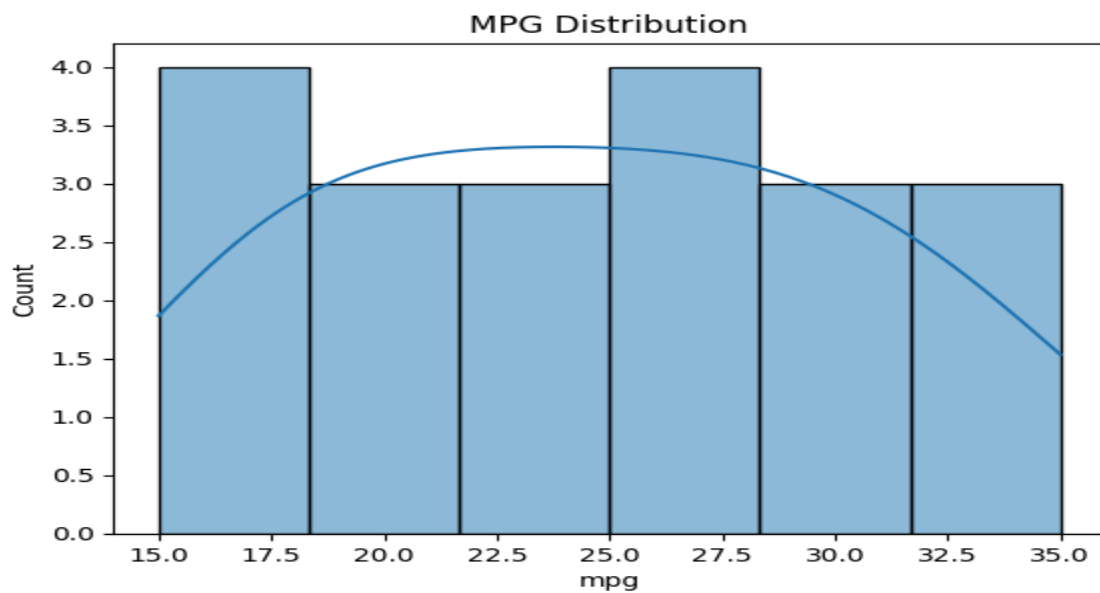
Output:

```

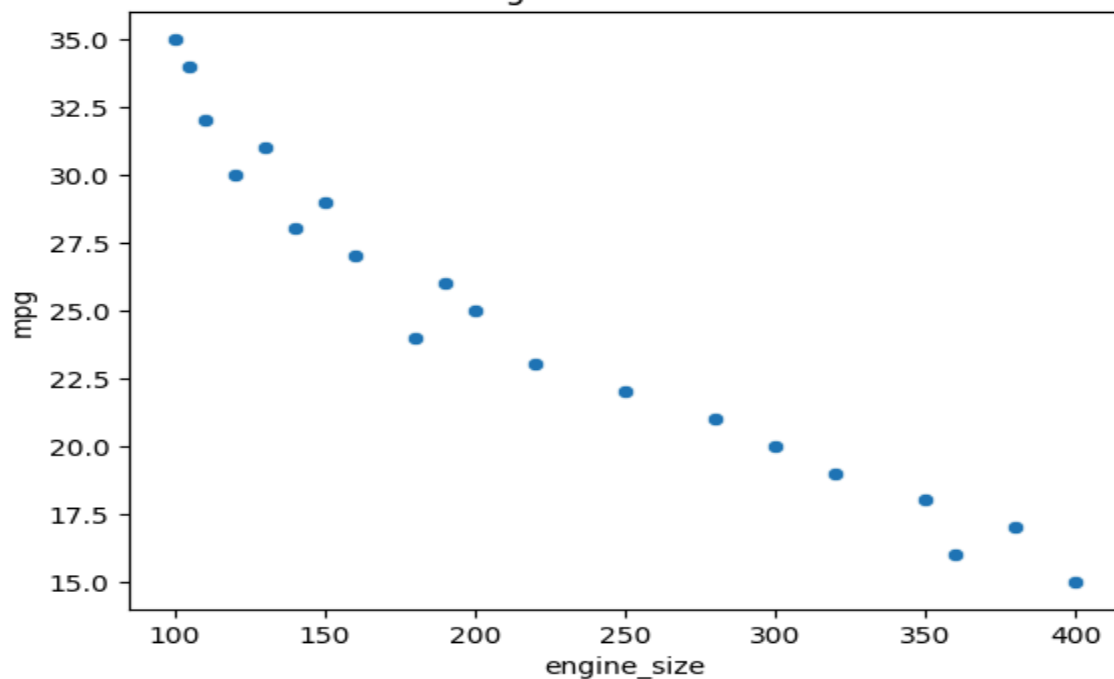
...   mpg  engine_size  weight  vehicle_type
0    18           350   3500         SUV
1    22           250   3200         Sedan
2    25           200   2800         Sedan
3    30           120   2200     Hatchback
4    27           160   2500         Sedan

      mpg  engine_size  weight
count  20.000000    20.000000    20.000000
mean   24.600000    222.250000   2910.000000
std     6.081897    100.242797    691.794006
min    15.000000    100.000000   1950.000000
25%    19.750000    137.500000   2375.000000
50%    24.500000    195.000000   2775.000000
75%    29.250000    305.000000   3425.000000
max    35.000000    400.000000   4200.000000

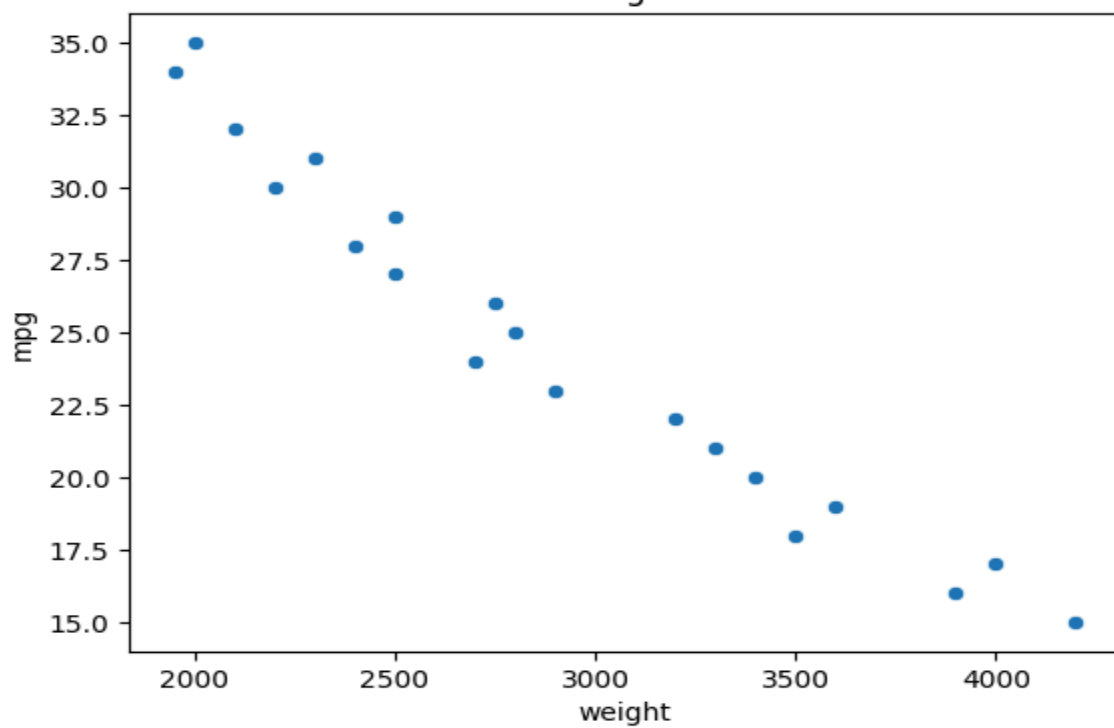
```

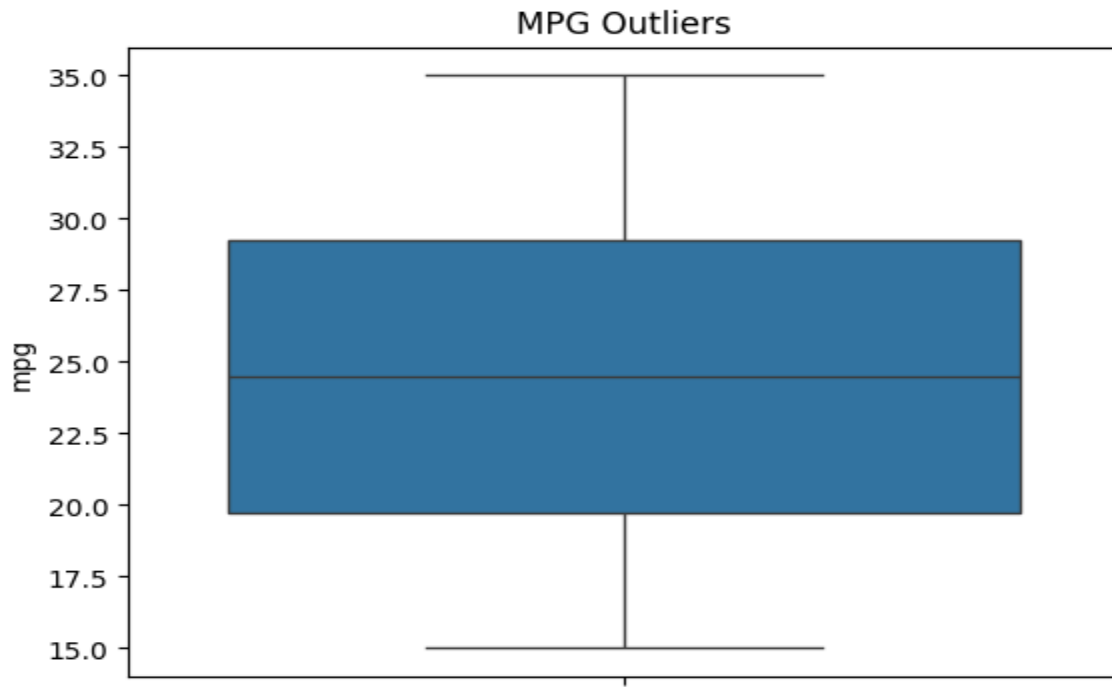


Engine Size vs MPG



Vehicle Weight vs MPG





Correlation:

	mpg	engine_size	weight
mpg	1.000000	-0.970073	-0.978472
engine_size	-0.970073	1.000000	0.991424
weight	-0.978472	0.991424	1.000000

Key Findings

- MPG shows the fuel-efficiency distribution of the vehicles.
- Vehicles with larger engines generally tend to have lower MPG.
- Heavier vehicles generally consume more fuel and have lower mileage.
- Different vehicle types show different fuel-efficiency levels.
- Boxplots help identify unusual MPG values and possible outliers.
- Correlation analysis helps understand the relationship between engine size, vehicle weight, and MPG.

Conclusion

The analysis shows that engine size and vehicle weight are important factors affecting fuel efficiency. Smaller and lighter vehicles generally provide better mileage, while larger and heavier vehicles tend to have lower MPG. Visualizations and correlation analysis help identify these patterns and detect unusual vehicle records.